

Exercise (2) – from Chapter 6. The complete list of problems is in the book.

Note: for these exercises, you can ignore non-ideal op-amp effects (e.g., bias currents).

1. For the circuit shown, find V_{out} as a function of V_{in} and determine the input impedance of the circuit.

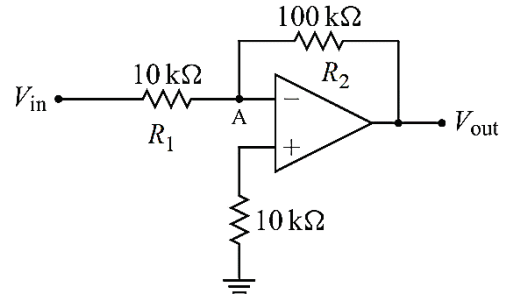
Solution:

This circuit is essentially an inverting amplifier. For an ideal op-amp, the resistor on the (+) input has no effect since there is no current through it and, thus, no voltage drop across it. Also, there is no current flowing into the (-) input. Hence, for node "A":

$$U_- = U_+ = 0 \text{ V}$$

$$I_{R1} + I_{R2} = 0 \rightarrow \frac{0 - V_{in}}{R_1} + \frac{0 - V_{out}}{R_2} = 0$$

$$V_{out} = -\frac{R_2}{R_1} V_{in} = -10V_{in}, \quad Z_{in} = \frac{V_{in}}{I_{in}} = R_1 = 10 \text{ k}\Omega$$



2. Derive an expression for the output voltage of the circuit in terms of the four input voltages. Simplify your result as much as possible.

Solution:

The circuit is redrawn with additional labels for clarity. Following the op-amp linear operation laws, for node B we get:

$$I_3 = \frac{V_3 - V_B}{R_3}, \quad I_4 = \frac{V_4 - V_B}{R_4}$$

$$V_B = I_0 R_0 = (I_3 + I_4) R_0 = \left(\frac{V_3 - V_B}{R_3} + \frac{V_4 - V_B}{R_4} \right) R_0$$

Rearranging this gives:

$$V_B \left(1 + \frac{R_0}{R_3} + \frac{R_0}{R_4} \right) = \left(\frac{V_3}{R_3} + \frac{V_4}{R_4} \right) R_0$$

Noting that all the resistors have the same value:

$$V_B \left(1 + \frac{R}{R} + \frac{R}{R} \right) = 3V_B = \left(\frac{V_3}{R} + \frac{V_4}{R} \right) R \rightarrow V_B = \frac{V_3 + V_4}{3}$$

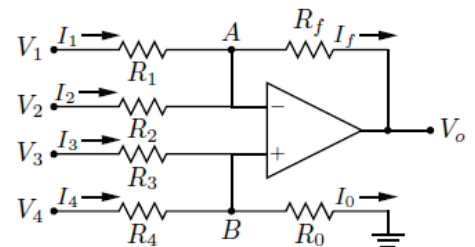
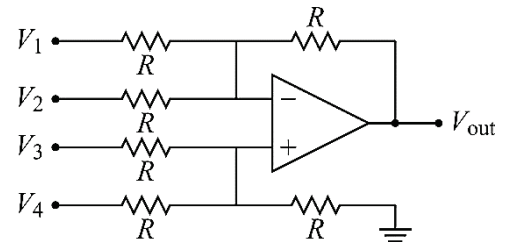
The output voltage is given by:

$$V_o = V_A - I_f R_f = V_A - (I_1 + I_2) R_f = V_A - \left(\frac{V_1 - V_A}{R_1} + \frac{V_2 - V_A}{R_2} \right) R_f$$

Noting that all the resistances are the same, and an ideal op-amp requires that $V_A = V_B$:

$$V_o = V_B - \left(\frac{V_1 - V_B}{R} + \frac{V_2 - V_B}{R} \right) R = 3V_B - V_1 - V_2 = 3 \frac{V_3 + V_4}{3} - (V_1 + V_2)$$

$$V_o = (V_3 + V_4) - (V_1 + V_2)$$



4. What is the maximum peak-to-peak output voltage for a 741 op-amp with ± 15 V power supplies? How small a load resistance will this amplifier drive with its output at the maximum level?

Solution:

Visiting the LM741 datasheet (TI), the relevant information is:

6.5 Electrical Characteristics, LM741⁽¹⁾

PARAMETER	TEST CONDITIONS		MIN	TYP	MAX	UNIT
Output voltage swing	$V_S = \pm 15$ V	$R_L \geq 10$ k Ω	± 12	± 14		V
		$R_L \geq 2$ k Ω	± 10	± 13		
Output short circuit current	$T_A = 25^\circ\text{C}$			25		mA

For loads $R_L > 10$ k Ω , $V_{\text{sat}}^\pm = \pm 15 \mp 1$ V = ± 14 V, the maximum output swing goes from -14 V to $+14$ V so:

$$V_{pp,\text{max}} = 28 \text{ V}$$

Since the output current of the LM741 is restricted to 25 mA and omitting the test condition for R_L , the minimum resistance it could drive is:

$$V_{\text{sat}} := 14 \text{ V}$$

$$I_{\text{max}} := 25 \text{ mA}$$

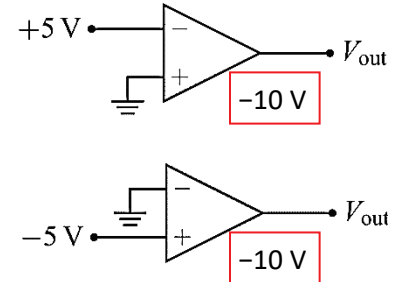
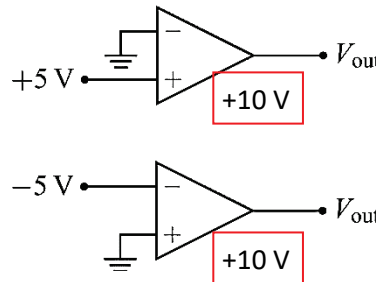
$$R_{\text{min}} := \frac{V_{\text{sat}}}{I_{\text{max}}} = 560 \text{ } \Omega$$

5. Refer to the figures and determine the output voltage for each circuit. Assume $V_{\text{sat}}^\pm = \pm 10$ V.

Solution:

All of these circuits are operating nonlinearly since there is no feedback connection, and the output is thus governed by:

$$V_{\text{out}} = A_{OL}(V_{\text{in}}^+ - V_{\text{in}}^-)$$



In each case, the magnitude of the voltage difference of the two inputs is 5 V and this will saturate the output at either ± 10 V (as $A_{OL} \gg 1$).

6. Draw the schematic diagram of an inverting amplifier with $R_i = 5$ k Ω and a voltage gain of -75 .

Solution:

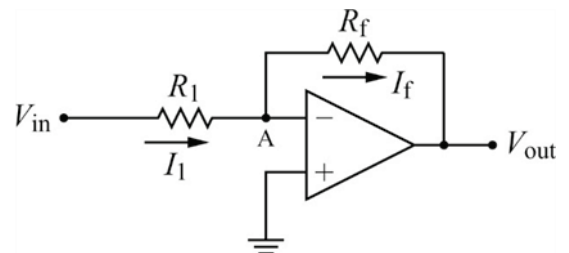
The circuit is the shown with parameters to be specified.
The input specification requires that:

$$R_i := R_1 = 5 \text{ k}\Omega$$

As the gain of the inverting amplifier is

$$A_V = -\frac{R_f}{R_1},$$

$$R_f := -A_V \cdot R_i = 375 \text{ k}\Omega$$



7. If the circuit of *Problem-6* uses ± 12 V power supplies, what input voltage will cause the output to saturate?

Solution:

For the inverting amplifier:

$$V_{\text{out}} = -\frac{R_f}{R_1} V_{\text{in}}$$

At saturation:

$$V_{\text{sat}} := -12 \text{ V} + 1 \text{ V} = -11 \text{ V}$$

$$V_{\text{in}} := \frac{V_{\text{sat}}}{A_V} = 0.147 \text{ V}$$

8. Determine the output voltage for each circuit shown in the figures.

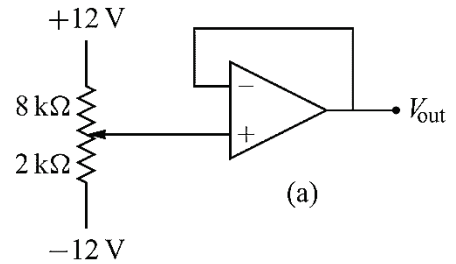
Solution:

- (a) The potentiometer acts like a **voltage divider** with 24 V across it. Since its bottom -12 V, we have to shift the result. The voltage at the (+) op-amp input is then:

$$V_{plus} := -12 \text{ V} + \frac{2 \text{ k}\Omega}{(2 + 8) \text{ k}\Omega} \cdot 24 \text{ V} = -7.2 \text{ V}$$

The op-amp is just a voltage follower, so:

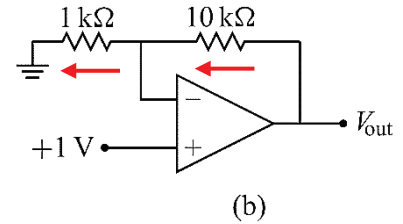
$$V_{out} := V_{plus} = -7.2 \text{ V}$$



- (b) By the first op-amp law, the voltage at the (-) input is 1 V. This produces a current of 1 mA through the 1 kΩ resistor going to the left. By the second law, all of this goes through the 10 kΩ resistor. KVL then gives:

$$V_{minus} := 1 \text{ V} \quad I := \frac{V_{minus}}{1 \text{ k}\Omega} = 1 \text{ mA}$$

$$V_{out} := V_{minus} + I \cdot 10 \text{ k}\Omega = 11 \text{ V}$$



- (d) This is a two-input **adder** circuit shown below. As before, the first op-amp law tells that $V_A = 0$. Ohm's law then gives the current through each of the input resistors, which then add by KCL:

$$I_1 = \frac{V_1}{R_1}, \quad I_2 = \frac{V_2}{R_2}, \quad I_f = I_1 + I_2,$$

since none of the current flows into the (-) input. Finally, KVL gives:

$$V_A - I_f R_f - V_{out} = 0 \rightarrow V_{out} = -I_f R_f = -\left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2\right)$$

Our circuit thus gives a **weighted sum** of the input voltages:

$$R_f := 1 \text{ k}\Omega \quad R_1 := 2 \text{ k}\Omega \quad R_2 := R_1 \quad V_1 := 1 \text{ V} \quad V_2 := 2 \text{ V}$$

$$V_{out} := -\left(\frac{R_f}{R_1} \cdot V_1 + \frac{R_f}{R_2} \cdot V_2\right) = -1.5 \text{ V}$$

